

# What to expect from logarithmic conformal field theory

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# Outline

- ① Rational vs logarithmic conformal field theory
- ②  $\beta\gamma$  ghosts as a logarithmic VOA

# Rational vertex operator algebras

- Module theory is semisimple.  
→ Virasoro  $L_0$  operator is diagonalisable.
- Finitely many inequivalent simple modules.
- Span of module characters carries a representation of the modular group  $\mathrm{SL}(2, \mathbb{Z})$ . [Zhu]
- Module category is a *modular tensor category*. [Verlinde, Moore-Seiberg, Huang]

Fusion product derived  
from correlation functions

=

Verlinde product derived  
from  $\mathrm{SL}(2, \mathbb{Z})$  action on  
characters

$$M_i \times M_j = \bigoplus_k N_{i,j}^k M_k,$$

$$N_{i,j}^k = \sum_l \frac{S_{i,l} S_{j,l} \overline{S_{k,l}}}{S_{0,l}}$$

# Logarithmic vertex operator algebras

- Module theory is **not** semisimple.  
→ Virasoro  $L_0$  operator can have **Jordan blocks**.
- Finitely many inequivalent simple modules if  $c_2$  condition is **satisfied**.
- Span of **torus 1-point functions** carries a representation of the modular group  $SL(2, \mathbb{Z})$ . [Miyamoto]
- Module category is **not** a *modular tensor category*. Instead just a braided tensor category [Huang-Lepowsky-Zhang] (provided tensor product exists).

**Verlinde formula must fail as characters can't distinguish all modules and modular action does not close on characters.**

# The $\beta\gamma$ ghost logarithmic conformal field theory

Two generating fields  $\beta$  and  $\gamma$ .

$$\gamma(z)\beta(w) \sim \frac{1}{z-w} \sim -\beta(z)\gamma(w), \quad \beta(z)\beta(w) \sim 0 \sim \gamma(z)\gamma(w)$$

$$T(z) = - : \beta(z) \partial \gamma(z) :, \quad J(z) = : \beta(z) \gamma(z) :.$$

Let  $\beta(z) = \sum_n \beta_n z^{-n-1}$  and  $\gamma(z) = \sum_n \gamma_n z^{-n}$  then

$$[\gamma_m, \beta_n] = \delta_{m+n,0} \mathbf{1}, \quad [\beta_m, \beta_n] = 0 = [\gamma_m, \gamma_n]$$

Triangular decomposition:

$$\mathfrak{G}' = \underbrace{\left( \bigoplus_{n \geq 1} \mathbb{C} \beta_{-n} \oplus \mathbb{C} \gamma_{-n} \right)}_{\mathfrak{n}_{-}} \oplus \mathbb{C} \gamma_0 \oplus \mathbb{C} \mathbf{1} \oplus \mathbb{C} \beta_0 \oplus \underbrace{\left( \bigoplus_{n \geq 1} \mathbb{C} \beta_n \oplus \mathbb{C} \gamma_n \right)}_{\mathfrak{n}_{+}}$$

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# Category $\mathcal{O}$

$$\mathfrak{G} = \underbrace{\left( \bigoplus_{n \geq 1} \mathbb{C}\beta_{-n} \oplus \mathbb{C}\gamma_{-n} \right)}_{\mathfrak{n}_-} \oplus \mathbb{C}\gamma_0 \oplus \underbrace{\mathbb{C}\mathbf{1} \oplus \mathbb{C}J_0 \oplus \mathbb{C}L_0}_{\mathfrak{h}} \oplus \mathbb{C}\beta_0 \oplus \underbrace{\left( \bigoplus_{n \geq 1} \mathbb{C}\beta_n \oplus \mathbb{C}\gamma_n \right)}_{\mathfrak{n}_+}$$

## Definition: Category $\mathcal{O}$

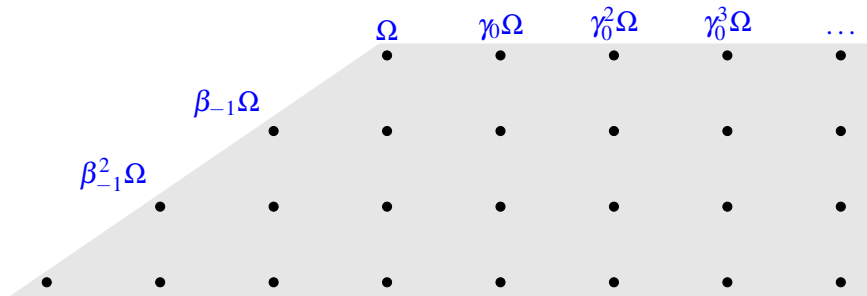
The category of  $\mathfrak{G}$  modules  $M$  satisfying

- $\mathbf{1}$  acts as identity operator.
- $M$  is finitely generated.
- $J_0$  acts semisimply ( $M$  is a weight module).
- Action of  $\mathfrak{n}_+$  is locally nilpotent:  $\forall m \in M$ ,  $U(\mathfrak{n}_+)m$  is finite-dimensional.

# Highest weight modules (category $\mathcal{O}$ )

## Proposition

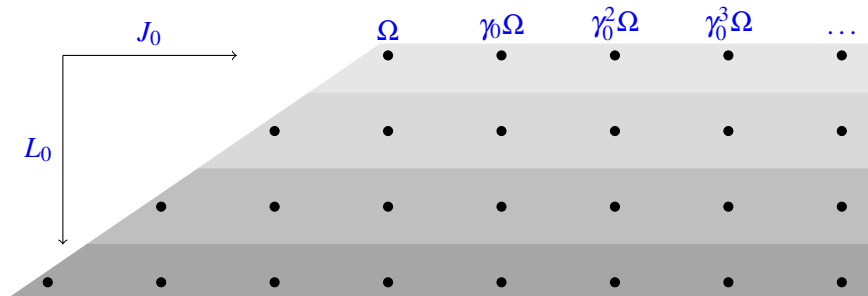
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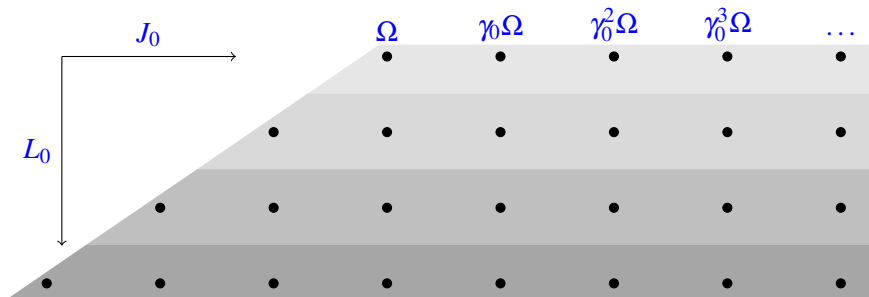
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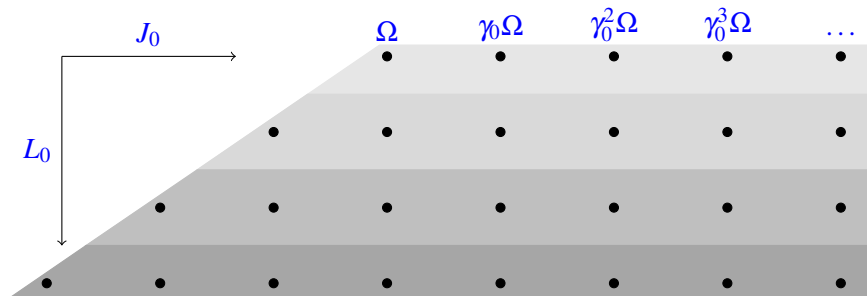


Closed under fusion,  $\mathcal{V} \times \mathcal{V} = \mathcal{V}$ , but not under action of  $\mathrm{SL}(2, \mathbb{Z})$ .

# Highest weight modules (category $\mathcal{O}$ )

## Proposition

$\mathcal{O}$  is semisimple. Contains only one simple object,  $\mathcal{V}$ , (up to equivalence).



$$\text{ch}[\mathcal{V}] = \text{tr}_{\mathcal{V}} z^{J_0} q^{L_0 - c/24} = \frac{q^{-1/12}}{\prod_{i=1}^{\infty} (1 - zq^i)(1 - z^{-1}q^{i-1})}, \quad |q| < 1, 1 < |z| < |q^{-1}|.$$

# New modules from automorphisms

$\mathfrak{G}$  admits two automorphisms

- Conjugation  $c(\beta_n) = \gamma_n$ ,  $c(\gamma_n) = -\beta_n$ ,  $c(J_0) = \mathbf{1} - J_0$ ,  $c(L_0) = L_0$ .
- Spectral flow  $\sigma^\ell(\beta_n) = \beta_{n-\ell}$ ,  $\sigma^\ell(\gamma_n) = \gamma_{n+\ell}$ ,  $\sigma^\ell(J_0) = J_0 + \ell \mathbf{1}$ ,  
 $\sigma^\ell(L_0) = L_0 - \ell J_0 - \frac{1}{2}\ell(\ell-1)\mathbf{1}$ ,  $\ell \in \mathbb{Z}$ .

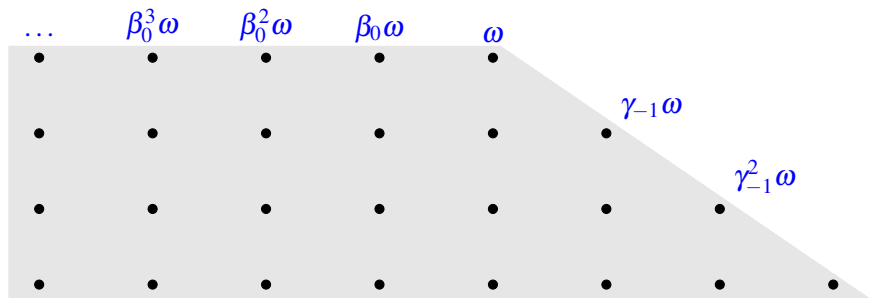
## Definition: Twisted module

Let  $M$  be a  $\mathfrak{G}$  module,  $\alpha$  an automorphism. Define  $\alpha$  twisted module by  $\alpha M \cong M$  as vector space. Twisted action  $x \cdot_\alpha m = \alpha^{-1}(x) \cdot m$ ,  $x \in \mathfrak{G}$ ,  $m \in M$ .

## Proposition

$c\mathcal{V} \cong \sigma^{-1}\mathcal{V}$  and  $\sigma^\ell\mathcal{V} \notin \mathcal{O}$  for  $\ell \neq 0$ .

# The module $c\mathcal{V}$



$$\text{ch}[c\mathcal{V}](z, q) \stackrel{!}{=} -\text{ch}[\mathcal{V}](z, q), \quad |q| < 1, |q| < |z| < 1.$$

# Going beyond highest weight modules

$$\mathfrak{G} = \underbrace{\left( \bigoplus_{n \geq 1} \mathbb{C}\beta_{-n} \oplus \mathbb{C}\gamma_{-n} \right)}_{\mathfrak{n}_{-}^p} \oplus \underbrace{\mathbb{C}\gamma_0 \oplus \mathbb{C}\mathbf{1} \oplus \mathbb{C}J_0 \oplus \mathbb{C}L_0 \oplus \mathbb{C}\beta_0}_{\mathfrak{h}^p} \oplus \underbrace{\left( \bigoplus_{n \geq 1} \mathbb{C}\beta_n \oplus \mathbb{C}\gamma_n \right)}_{\mathfrak{n}_{+}^p}$$

## Definition: Category $\mathcal{R}$

The category of  $\mathfrak{G}$  modules  $M$  satisfying

- $\mathbf{1}$  acts as identity operator.
- $M$  is finitely generated.
- $J_0$  acts semisimply ( $M$  is a weight module).
- Action of  $\mathfrak{n}_{+}^p$  is locally nilpotent:  $\forall m \in M$ ,  $U(\mathfrak{n}_{+}^p)m$  is finite-dimensional.

Simple objects of  $\mathcal{R}$  can be induced from simple  $\mathfrak{h}^p$  modules.

# $\mathfrak{h}^p$ weight module classification

## Theorem [Block]

Any simple  $\mathfrak{h}^p$  weight module is equivalent to one of the following.

- 1 Unique highest weight module:  $\overline{\mathcal{V}} = \mathbb{C}[\gamma_0]\overline{\Omega}$ ,  $\beta_0\overline{\Omega} = 0$ .  $\rightarrow J_0\overline{\Omega} = 0$ .
- 2 Unique lowest weight module:  $\overline{\mathcal{V}}^* = \mathbb{C}[\beta_0]\overline{\omega}$ ,  $\gamma_0\overline{\omega} = 0$ .  $\rightarrow J_0\overline{\omega} = \overline{\omega}$ .
- 3 Dense module: For  $\lambda \in \mathbb{C} \setminus \mathbb{Z}$ , let

$$\overline{\mathcal{W}}_\lambda = \mathbb{C}[\beta_0]\beta_0 u_\lambda \oplus u_\lambda \oplus \mathbb{C}[\gamma_0]\gamma_0 u_\lambda$$

be the module generated by a weight vector  $u_\lambda$ ,  $J_0 u_\lambda = \lambda u_\lambda$ .  
 $\overline{\mathcal{W}}_\lambda \cong \overline{\mathcal{W}}_\mu$  iff  $\lambda - \mu \in \mathbb{Z}$ .

For  $\lambda = 0$ , there exist two indecomposables characterised by the non-split exact sequences

$$0 \longrightarrow \overline{\mathcal{V}} \longrightarrow \overline{\mathcal{W}}_0^+ \longrightarrow \overline{\mathcal{V}}^* \longrightarrow 0, \quad 0 \longrightarrow \overline{\mathcal{V}}^* \longrightarrow \overline{\mathcal{W}}_0^- \longrightarrow \overline{\mathcal{V}} \longrightarrow 0.$$

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# Going beyond highest weight modules

$\mathcal{V}$  and  $c\mathcal{V}$  are the inductions of  $\overline{\mathcal{V}}$  and  $\overline{\mathcal{V}}^*$ . Let  $\mathcal{W}_\lambda$  and  $\mathcal{W}_0^\pm$  be the inductions of  $\overline{\mathcal{W}}_\lambda$  and  $\overline{\mathcal{W}}_0^\pm$ .

## Theorem

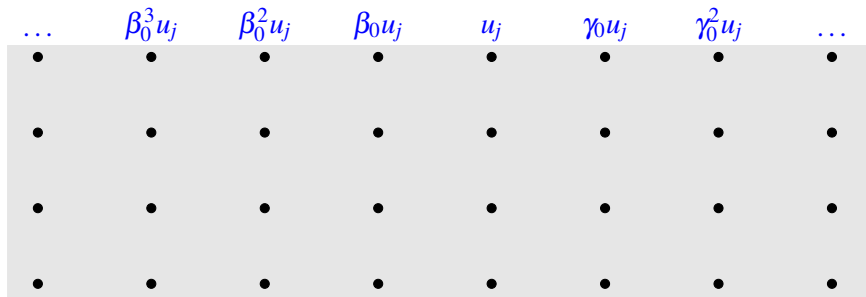
Up to isomorphism, the simple objects of  $\mathcal{R}$  are  $\mathcal{V}$ ,  $c\mathcal{V}$ ,  $\mathcal{W}_\lambda$ ,  $\lambda \in \mathbb{C}$ ,  $\operatorname{re}(\lambda) \in (0, 1)$ .

Note:  $\sigma^\ell \mathcal{W}_\lambda \notin \mathcal{R}$  for  $\ell \neq 0$ ,  $\sigma^\ell \mathcal{V} \notin \mathcal{R}$  for  $\ell \neq 0, -1$ .

## Definition

Let  $\mathcal{T}$  be the module category generated by  $\mathcal{V}$ ,  $\mathcal{W}_\lambda$ ,  $\lambda \in (0, 1)$ , spectral flow twists and extensions.

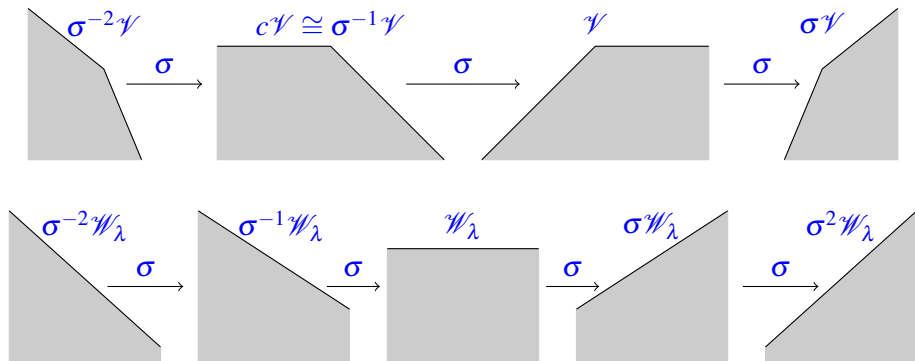
# The $\mathscr{W}_\lambda$ modules



$$\text{ch}[\mathscr{W}_\lambda](z, q) = \frac{z^\lambda q^{-1/12}}{\prod_{i=1}^{\infty} (1 - q^i)^2} \sum_{m \in \mathbb{Z}} z^m$$

# Spectral flow twists

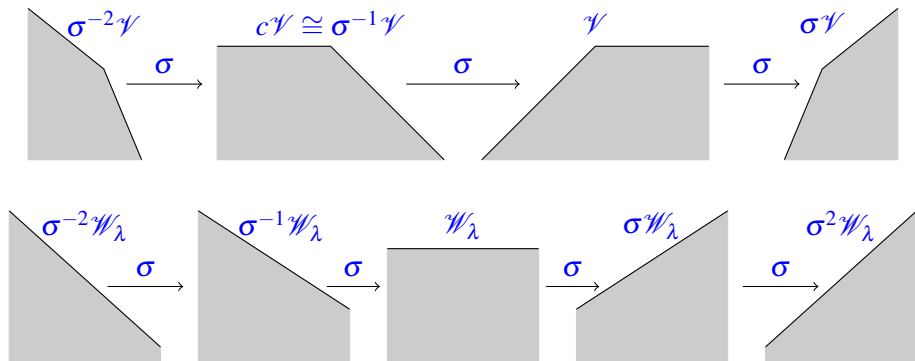
Spectral flow 'tilts' the energy grading:



The action of the modular group closes on the span of the characters of  $\sigma^\ell \mathscr{W}_\lambda$ ,  $\ell \in \mathbb{Z}$ ,  $\lambda \in (0, 1)$ .

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The action of the modular group closes on the span of the characters of  $\sigma^\ell \mathcal{W}_\lambda$ ,  $\ell \in \mathbb{Z}$ ,  $\lambda \in (0, 1)$ .

# Towards a logarithmic Verlinde formula

The action of the modular group closes on the span of  $\sigma^{\ell\mathscr{W}}_{\lambda}$  characters.

$$\begin{aligned} \mathbf{S} : \text{ch} \left[ \sigma^{\ell\mathscr{W}}_{\lambda} \right] &\mapsto \sum_{m \in \mathbb{Z}} \int_0^1 \mathbf{S} \left[ \sigma^{\ell\mathscr{W}}_{\lambda} \rightarrow \sigma^{m\mathscr{W}}_{\mu} \right] \text{ch} \left[ \sigma^{m\mathscr{W}}_{\mu} \right] d\mu \\ \mathbf{S} \left[ \sigma^{\ell\mathscr{W}}_{\lambda} \rightarrow \sigma^{m\mathscr{W}}_{\mu} \right] &= (-1)^{\ell+m} e^{-2\pi i(\ell\mu+m\lambda)} \end{aligned}$$

This  $\mathbf{S}$  matrix is symmetric and unitary.

What about  $\sigma^{\ell\gamma}$ ?

# Towards a logarithmic Verlinde formula

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$$S \left[ \sigma^{\ell\mathcal{W}}_{\lambda} \rightarrow \sigma^{m\mathcal{W}}_{\mu} \right] = (-1)^{\ell+m} e^{-2\pi i(\ell\mu+m\lambda)}$$

This  $S$  matrix is symmetric and unitary.

What about  $\sigma^{\ell\mathcal{V}}$ ?

# Towards a logarithmic Verlinde formula

Recall,  $0 \longrightarrow \sigma^{-1}\mathcal{V} \longrightarrow \mathcal{W}_0^- \longrightarrow \mathcal{V} \longrightarrow 0$ .

We can resolve  $\mathcal{V}$  in terms of the  $\sigma^\ell \mathcal{W}_0^-$  by splicing exact sequences.

$$\cdots \longrightarrow \sigma^{-2}\mathcal{W}_0^- \longrightarrow \sigma^{-1}\mathcal{W}_0^- \longrightarrow \mathcal{W}_0^- \longrightarrow \mathcal{V} \longrightarrow 0$$

The character of  $\mathcal{V}$  can then be computed using the Euler-Poincaré principle.

$$\mathrm{ch} \left[ \sigma^\ell \mathcal{V} \right] = \sum_{m=0}^{\infty} (-1)^m \mathrm{ch} \left[ \sigma^{\ell-m} \mathcal{W}_0^- \right].$$

S-transformation:

$$\mathrm{S} \left[ \sigma^\ell \mathcal{V} \rightarrow \sigma^m \mathcal{W}_\mu \right] = (-1)^{\ell+m+1} \frac{e^{-2\pi i(\ell+1/2)\mu}}{e^{\pi i\mu} - e^{-\pi i\mu}}$$

## Recall the rational Verlinde formula

Rational vertex operator algebras admit only a finite number of inequivalent simple modules  $M_i$ ,  $i = 0, \dots, n$ , where  $M_0$  is the vacuum module.

Fusion:  $M_i \times M_j = \bigoplus_k N_{i,j}^k M_k$ ,  $N_{i,j}^k \in \mathbb{N}_0$ .

Action of modular group closes on span of module characters:

$$T : \text{ch}[M_i] \mapsto T_i \text{ch}[M_i]$$

$$S : \text{ch}[M_i] \mapsto S_{i,j} \text{ch}[M_j]$$

The Verlinde formula relates the fusion structure constants and the  $S$ -matrix coefficients.

$$N_{i,j}^k = \sum_n \frac{S_{i,n} S_{j,n} \overline{S_{k,n}}}{S_{0,n}}$$

# Towards a logarithmic Verlinde formula

Natural proposal for a logarithmic Verlinde product:

$$\mathrm{ch}[M \times N] = \sum_{m \in \mathbb{Z}} \int_0^1 N_{M,N}^{m,\mu} \mathrm{ch}[\sigma^m \mathcal{W}_\mu] \, \mathrm{d}\mu.$$

$$N_{M,N}^{m,\mu} = \sum_{n \in \mathbb{Z}} \int_0^1 \frac{\mathrm{S}[M \rightarrow \sigma^n \mathcal{W}_v] \mathrm{S}[N \rightarrow \sigma^n \mathcal{W}_v] \overline{\mathrm{S}[\sigma^m \mathcal{W}_\mu \rightarrow \sigma^n \mathcal{W}_v]}}{\mathrm{S}[\mathcal{V} \rightarrow \sigma^n \mathcal{W}_v]} \, \mathrm{d}v$$

# Towards a logarithmic Verlinde formula

## Theorem

Verlinde product satisfies

$$\begin{aligned}\mathrm{ch}[cM \times cN] &= \mathrm{ch}[c(M \times N)] \\ \mathrm{ch}[\sigma^\ell M \times \sigma^k N] &= \mathrm{ch}[\sigma^{\ell+k}(M \times N)]\end{aligned}$$

and

$$\begin{aligned}\mathrm{ch}[\mathcal{V} \times \mathcal{V}] &= \mathrm{ch}[\mathcal{V}], \\ \mathrm{ch}[\mathcal{V} \times \mathcal{W}_\mu] &= \mathrm{ch}[\mathcal{W}_\mu], \\ \mathrm{ch}[\mathcal{W}_\lambda \times \mathcal{W}_\mu] &= \mathrm{ch}[\mathcal{W}_{\lambda+\mu}] + \mathrm{ch}[\sigma^{-1}\mathcal{W}_{\lambda+\mu}].\end{aligned}$$

Verlinde product and fusion product agree in spot checks.

Fusion products uniquely determined by characters unless  $\lambda + \mu \in \mathbb{Z}$ .

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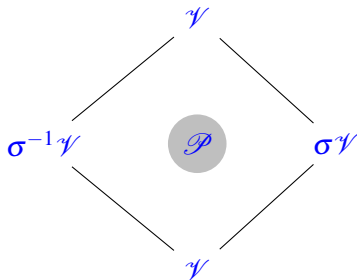
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# Indecomposable fusion products

By direct computation:  $\mathcal{W}_\lambda \times \mathcal{W}_{-\lambda} = \sigma^{-1} \mathcal{P}$ , where



$$0 \longrightarrow \mathcal{W}_0^+ \longrightarrow \mathcal{P} \longrightarrow \sigma \mathcal{W}_0^+ \longrightarrow 0, \quad 0 \longrightarrow \sigma \mathcal{W}_0^- \longrightarrow \mathcal{P} \longrightarrow \mathcal{W}_0^- \longrightarrow 0.$$

## Conjecture

In  $\mathcal{T}$

- the  $\sigma^{\ell} \mathcal{W}_{\lambda}$ ,  $\lambda \in (0, 1)$ , are simple and projective,
- the  $\sigma^{\ell} \mathcal{P}$  are indecomposable projective covers of  $\sigma^{\ell} \mathcal{V}$ ,
- the Verlinde product matches the fusion product (at the level of characters).

# General pattern

- 1 Preferred class of modules, called standard modules, parametrised by a measurable space  $(M, \mu)$ .
- 2 The subset of  $M$  corresponding to non-simple standard modules has measure zero with respect to  $\mu$ .
- 3 Simple standard modules (conjecturally) admit no extensions.
- 4 The characters of the standard modules form a (topological) basis for the space of all characters.
- 5 The characters of the standard modules span a representation of the modular group  $SL(2, \mathbb{Z})$ . In the standard basis, the  $S$ -transformation defines a symmetric, unitary operator whose square may be identified with conjugation.
- 6 Combining the  $S$ -transformation with the obvious generalisation of the standard Verlinde formula leads to fusion rules at the level of characters (with non-negative integer coefficients).

# General pattern

The standard module pattern has been observed for:

- The free boson
- The  $(1, p)$  and  $(p, q)$  singlet algebras [Creutzig-Milas, Ridout-SW, Creutzig-Milas-SW]
- The  $\beta\gamma$  ghosts [Ridout-SW]
- Affine  $\mathfrak{sl}(2)$  [Creutzig-Ridout]
- Affine  $\mathfrak{osp}(1|2)$  [Ridout-Snadden-SW]
- and more...